

Reflective Journal

Reflective Journal: Module 09 Lab – Ensemble Methods

For Module 09, I learned about ensemble methods and how using multiple models together can sometimes give better and more stable results than using only one model. Before this lab, I understood the basic idea of a Decision Tree, but I did not really understand why a Random Forest would be better. Working through the lab made it more clear because I could directly compare a single Decision Tree with a Random Forest.

In the first part of the lab, I trained one Decision Tree using the Iris dataset. The model ended up getting 100% accuracy on the test data. After that, I trained a Random Forest using 100 trees, and it also got 100% accuracy. At first, I expected the Random Forest to automatically score higher because it uses many trees, but this lab showed me that a more advanced model will not always have a higher score on every dataset. Since the Iris dataset is small and the classes are separated pretty clearly, both models were able to perform well.

The biggest thing I learned is why Random Forest is considered more stable than a single Decision Tree. A single tree can change a lot if the training data changes even a little. Random Forest helps with that by training many different trees on random samples of the data and then combining their answers. Even though the accuracy was the same in this lab, I can see why the Random Forest would probably be more reliable on a larger or more complicated dataset.

I also liked the feature-importance part of the lab because it gave me a better idea of what the model was actually using to make its predictions. Petal width and petal length were the two most important features, while the sepal measurements mattered less. That result made sense because the petal measurements seem to separate the different flower species more clearly. The graph made the result easier to understand than just looking at the numbers.

One part that confused me at first was the difference between bagging and boosting. After reviewing the module, I understood that Random Forest is a bagging method because the trees are trained independently and their predictions are combined. Boosting is different because each new model tries to fix mistakes made by the earlier models. I still need more practice with boosting, but I understand the main difference better now.

Overall, this lab helped me understand why ensemble methods are useful. My main takeaway is that combining several models can reduce the weaknesses of relying on one model alone. Even though both models had the same accuracy in this lab, the Random Forest is still designed to give more stable predictions by using the combined vote of many trees.